



2-Dec-2024

## **LIMITED SOURCES JUSTIFICATION**

### **1. Nature and/or Description of Action**

The Defense Threat Reduction Agency contracting activity proposes to restrict consideration of schedule contractors to fewer than required in FAR 8.405-2 to procure an effort entitled Arms Control Subject Matter Expert (SME) Support, particularly to support an independent design and development review of the next-generation Radionuclide Aerosol Sampler/Analyzer (RASA), a specialized system used in the International Monitoring System (IMS). The planned performer is Desert Research Institute (DRI), a subsidiary of the University of Nevada system. This requirement will be awarded as a sole source contract under the General Services Administration (GSA) Multiple Award Schedule (MAS).

### **2. Description of the Supplies/Services Required**

The Defense Threat Reduction Agency (DTRA) executes and provides mission support to multiple treaties, agreements, and treaty provisional secretariats to which the United States is a party or other type of participant. Such treaties include the New Strategic Arms Reduction Treaty (START), International Atomic Energy Agency (IAEA) agreements, the Chemical Weapons Convention (CWC), and the Provisional Secretariat of the Comprehensive Nuclear-Test-Ban Treaty (CTBT). Due to the constantly evolving nature of treaty policy, ongoing Research, Development, Test and Evaluation (RDTE) is required to ensure that the United States is prepared to execute inspection, verification, and validation to support its mission and associated international engagements.

The CTBT verification regime includes the International Monitoring System (IMS). DTRA currently operates and manages 31 U.S. IMS stations. The continuation of U.S. operation and RDTE investments into the U.S. IMS stations provides access to data from IMS stations worldwide and supports U.S. nuclear explosion monitoring and detection objectives. Detecting nuclear debris from a nuclear weapon exploded in or substantially vented to the Earth's atmosphere provides an indication that a violation of the CTBT has occurred. For this reason, radionuclide particulate samplers in the IMS sample large volumes of air each day and make sensitive gamma-ray spectroscopy measurements of the samples collected. An automated system, the Radionuclide Aerosol Sampler/Analyzer (RASA), was developed 30 years ago at Pacific Northwest National Laboratory (PNNL) to meet CTBT aerosol measurement requirements. This RASA consists of a novel sampling apparatus, a high-resolution germanium detector, and very sophisticated software. One of the original developers, designers, testers, and implementers of the original RASA currently works at DRI and provides a unique capability regarding the understanding and importance of radionuclide particulate samplers and their role in the monitoring regime. Their highly specialized knowledge and unique experience into the previous RASA and insight for improvements in a next-generation RASA cannot be replicated through sources in the commercial sector.

DTRA seeks to leverage its unique capabilities and expertise in arms control implementation, nuclear deterrence support, and research and development to support these goals. The scope of this requirement includes RDTE of equipment, methodologies, and other support essential to DTRA's mission to focus on emerging and trending threats in arms control. DTRA seeks to develop and test the viability of technological solutions and operational concepts to support a whole-of-government approach to limit, reduce, or mitigate such threats.

DTRA requires specialized subject matter expert (SME) support in all of the following areas:

- Advising DTRA on the technical aspects of the next-generation (RASA)
- Documenting how the new system will meet IMS and DTRA requirements
- Communicating with and for DTRA on the scientific implications on verification and monitoring.
- Senior scientist expertise (Ph.D.-level) in nuclear physics, nuclear engineering, or similar with a demonstrated background in radiation detection and trace radionuclide measurements. The expert should have a minimum of 20 years of demonstrated experience in all of the following area:
  - Project management on Federal R&D projects
  - Leadership of scientific groups
  - Treaty verification/nuclear monitoring of the environment
  - Direct technical development of the RASA
  - Speaking, writing science reports, and publishing in journals relevant to the technical topic
  - Senior Scientist to provide expert-level inputs to develop R&D for next-generation arms control efforts, to include demonstrated experience in monitoring and verification equipment and treaty verification, authentication, and compliance activities.

This project will consist of a 12-month period of performance (PoP), one 12-month option, and two surge options that may be executed in concurrence with the base and option POPs, for a maximum 24-month PoP. The total estimated value of this SME technical support effort is no more than \$1.1M.

### **3. Statutory Authority**

FAR 8.405-6(a)(1)(i)(B) - only one source is capable of providing the supplies or services required at the level of quality required because the supplies or services are unique or highly specialized.

### **4. Justification Rationale**

This project requires highly specialized knowledge and unique experience with the original RASA and knowledge of efforts to develop a next-generation RASA; which limits the pool of available performers who are able to complete this requirement. The RASA knowledge and support to be provided is unique to the international treaty monitoring mission of the IMS and applicability is limited throughout the commercial sector. Due to the specialized nature of the equipment, which is for Government treaty purposes, this type of work is unlikely to be undertaken by the commercial sector without significant government investment. The expertise and experience that DRI provides is unique in that DRI retains personnel with unique technical and historical expertise required to support the development of the next-generation RASA. The RASA was developed 30 years ago at PNNL to meet CTBT aerosol measurement requirements. One of the original developers, designers, testers, and implementers of the original RASA

currently works at DRI and provides a unique capability regarding the understanding and importance of radionuclide particulate samplers and their role in the monitoring regime. Their highly specialized knowledge and unique experience into the previous RASA and insight for improvements in a next-generation RASA cannot be replicated through sources in the commercial sector. Finally, DRI has the specialized operational knowledge and experience necessary to ensure that DTRA meets the requirements of the Comprehensive Nuclear-Test-Ban Treaty Organization's (CTBTO's) Provisional Secretariat.

Based on the Government's technical needs and market research, the required services are highly specialized and unique to meet the Government's technical requirements. Therefore, DRI is the only performer that satisfies the criteria necessary for successful performance of this effort.

## **5. Market Research**

In accordance with FAR Part 10, formal market research has been conducted for this acquisition. The research sources used consisted of a sources-sought notice, prior acquisition information, a review of government and commercial databases, and internet resources. The market research found no sources capable of reasonably meeting this requirement other than DRI.

A sources-sought announcement "HDTRA1ArmsControlSMESPT" was posted from June 6 through July 8, 2024 which resulted in six (6) responses. The government reviewed each offerors response and determined that information provided were either non-responsive or lacked the experience required to perform the requirements of the contract.

Based on market research and procurement history, there is no other performer than DRI who can provide the expertise required for this project. DRI currently retains the RASA expertise that formerly resided at PNNL over the past three decades.

## **6. Other Supporting Facts**

Only one other performer (the Pacific Northwest National Laboratory (PNNL)) has experience with the RASA. Since PNNL is participating in the design of the next-generation RASA, they cannot provide the independent expertise required to assess the system. The expertise that DRI can provide is required to ensure that the investment required in the next-generation RASA (several million dollars) would be technically justifiable and would meet the Government's and CTBTO's requirements.

## **7. Subsequent Actions**

DTRA will continue to conduct market surveillance through-out the performance of this follow-on effort.

### **Technical Certification**

I certify that the data and information forming the basis for this justification are accurate and complete to the best of my knowledge and belief.

[Redacted Signature Area]

\_\_\_\_\_  
Program Manager/Project Officer

\_\_\_\_\_  
Date

### **Contracting Officer Certification and Determination**

I certify that this justification is accurate and complete to the best of my knowledge and belief. Additionally, I determine that the order represents the best value consistent with FAR Subpart 8.404(d).

[Redacted Signature Area]

\_\_\_\_\_  
Contracting Officer

\_\_\_\_\_  
Date

### **General Counsel Coordination**

This justification is legally sufficient.

[Redacted Signature Area]

\_\_\_\_\_  
Attorney

\_\_\_\_\_  
Date

### **Approval**

Based on the above justification, I hereby approve accomplishment of this acquisition by means of limited sources.

[Redacted Signature Area]

\_\_\_\_\_  
Competition Advocate

\_\_\_\_\_  
Date